Functions

A function is a special type of relation. Every function is a relation. But not every relation is a function.

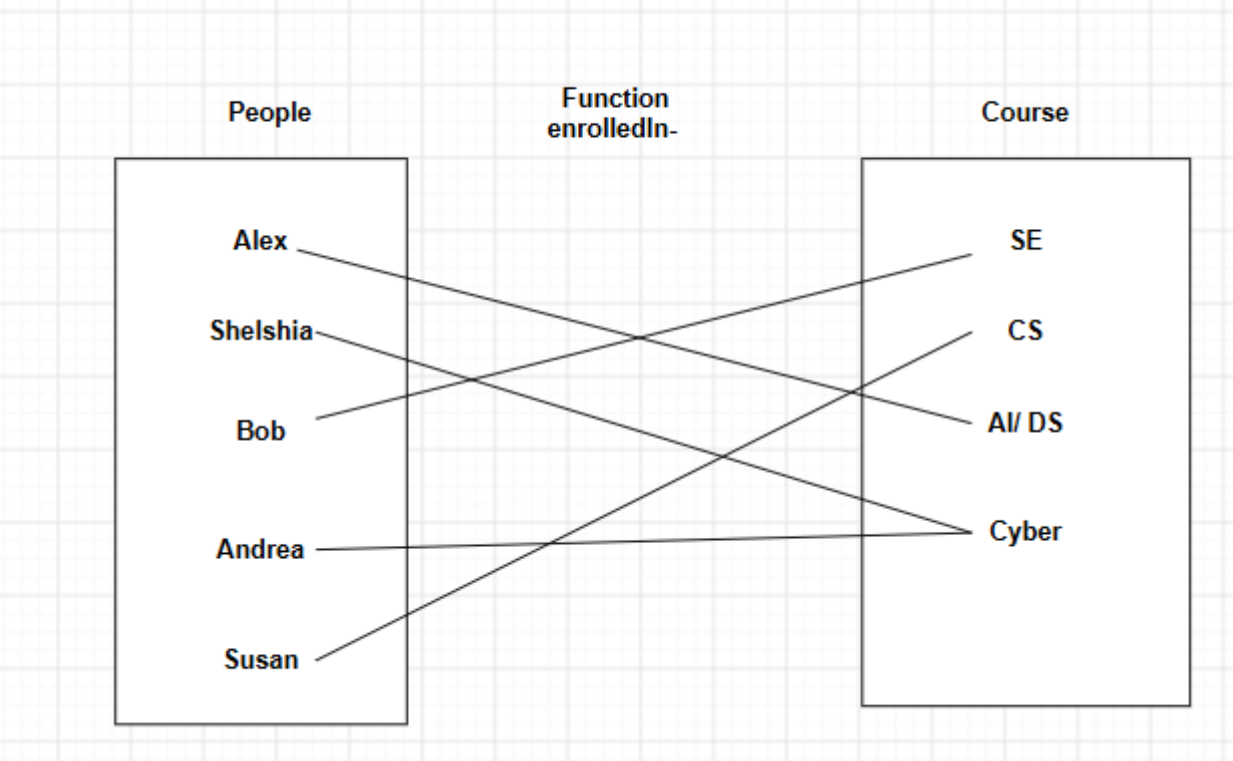
Considering a relation, a relation can have

1. One to One
2. One to Many
3. Many to One
4. Many to Many

Considering Function, Function can only have

1. One to One
2. Many to One

For a relation to be a function it cannot have one to many or many to many



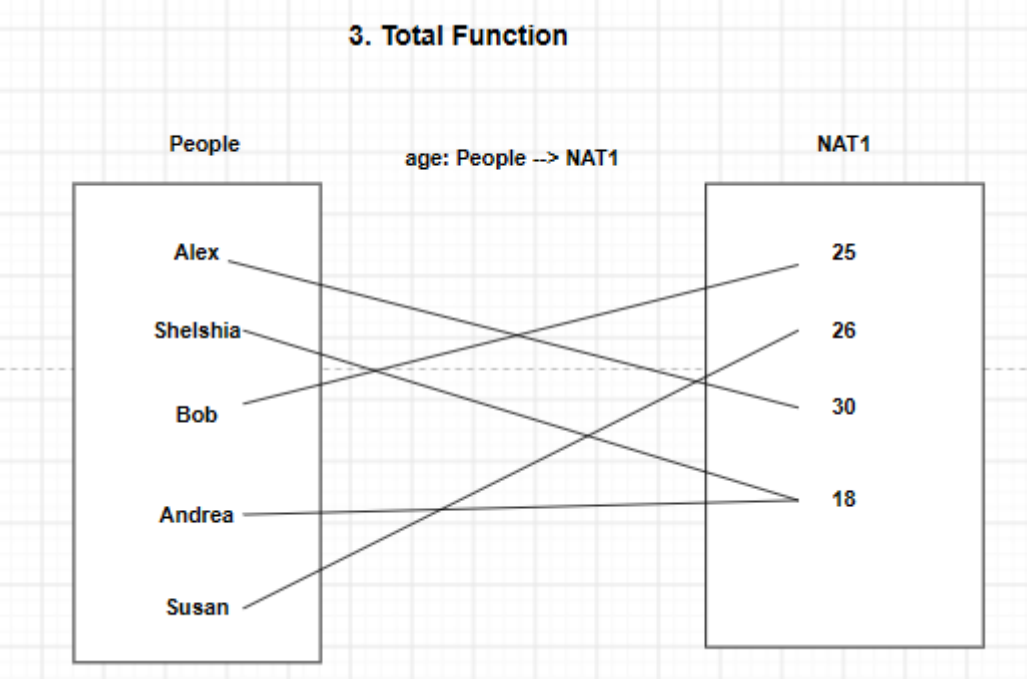
A function can be of 2 types

1. **Total Function**
2. **Partial Function**

Total Function- All elements of source set is having a pair

Example age: People → NAT1

Every element in source set will have a pair. If we take age every person in people set will have a pair.

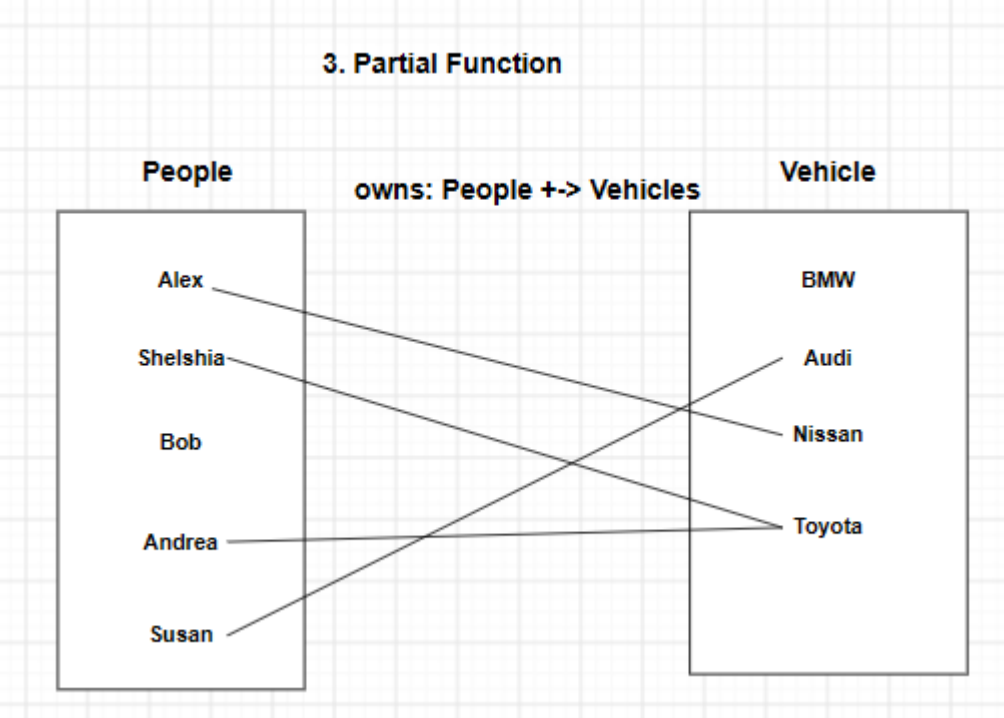


Partial Function

Not all elements of source set will have a pair

Ex- owns : People +-> Vehicle

Not every person will own a vehicle



Additionally there are functional classes

Injective- One to One relationship

Surjective- All elements of target set will have a pair.

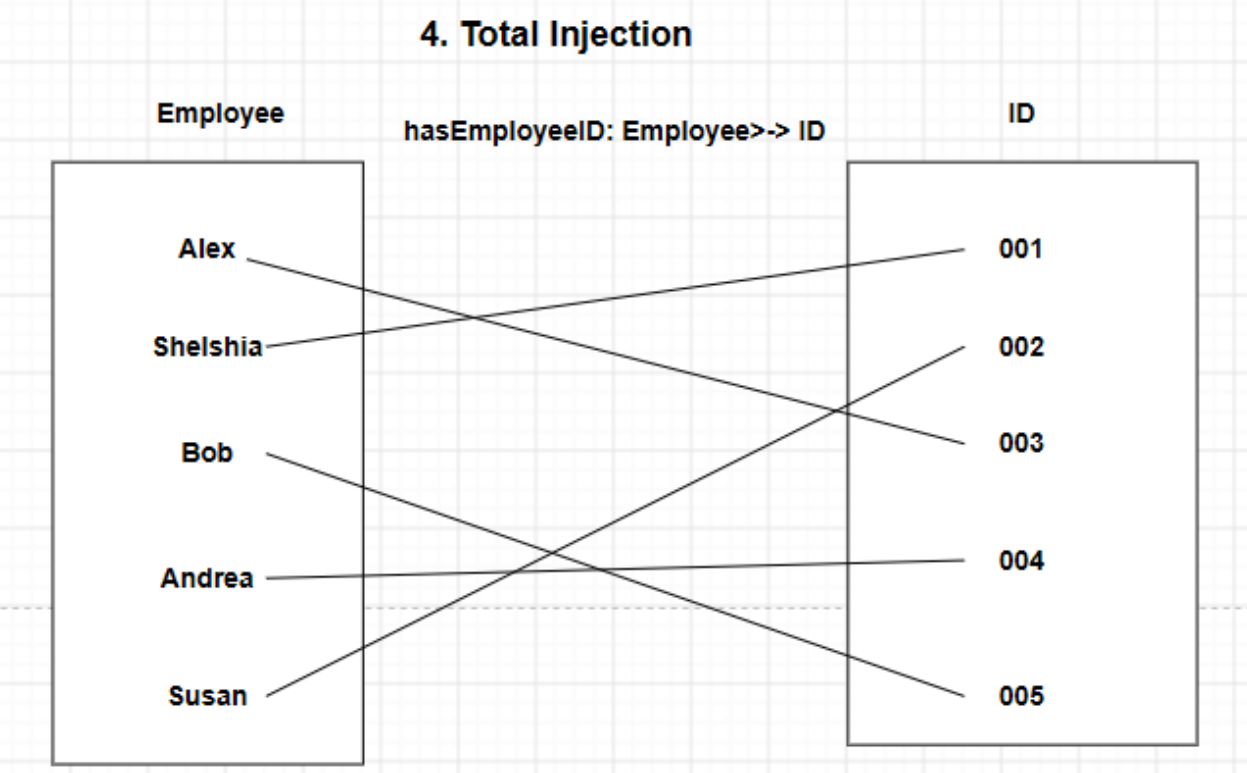
Bijective is a combination of both injective and surjective.

1. **Total Injection**

Total Injection means all elements in source set is having a pair and is having one to one relationship

Ex- hasEmployeeID : Employee >-> ID

Every employee will have an employee id and it will be unique to each employee



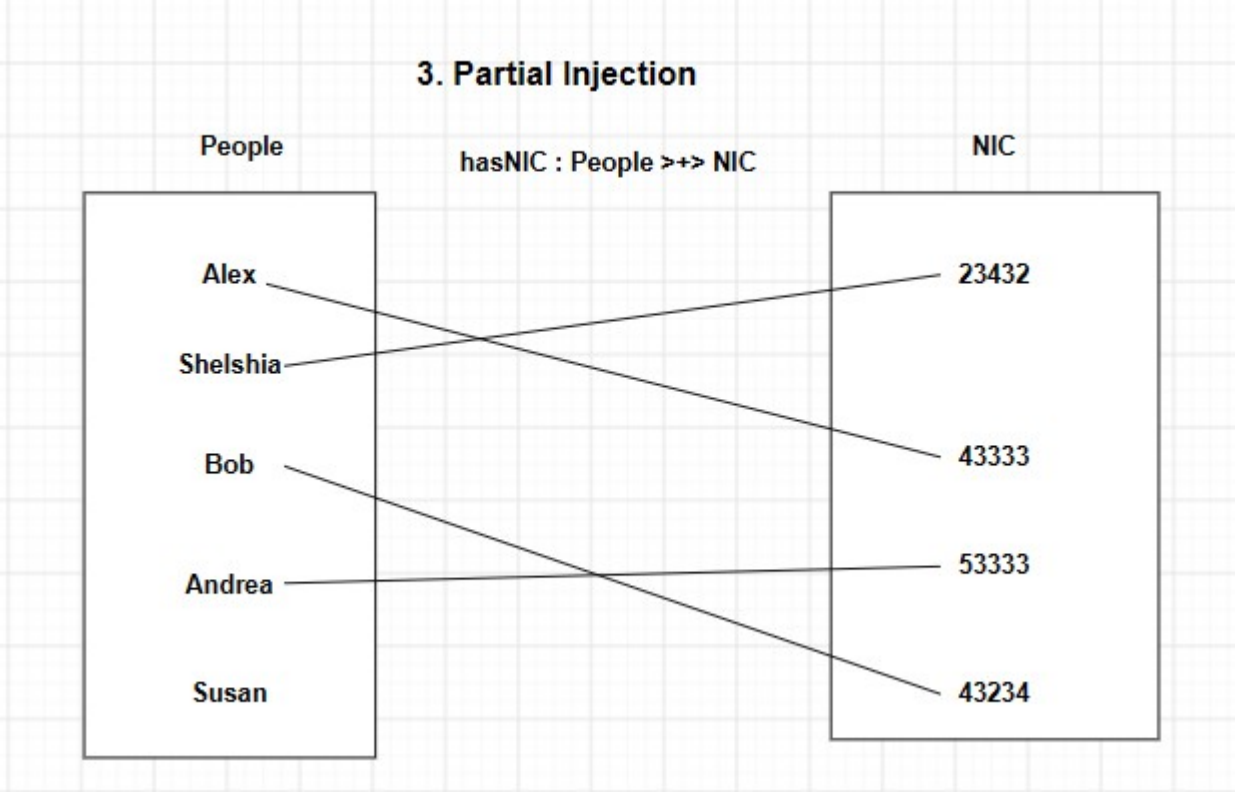
All employees have an unique id. Therefore, it's a total injection

2. **Partial Injection**- Not all elements from source set will have a pair. However, the ones that have pairs will have one to one pair

Ex- hasNIC

Not every individual has NIC. Only people above 15 will have NIC.

hasNIC: People >+> NIC

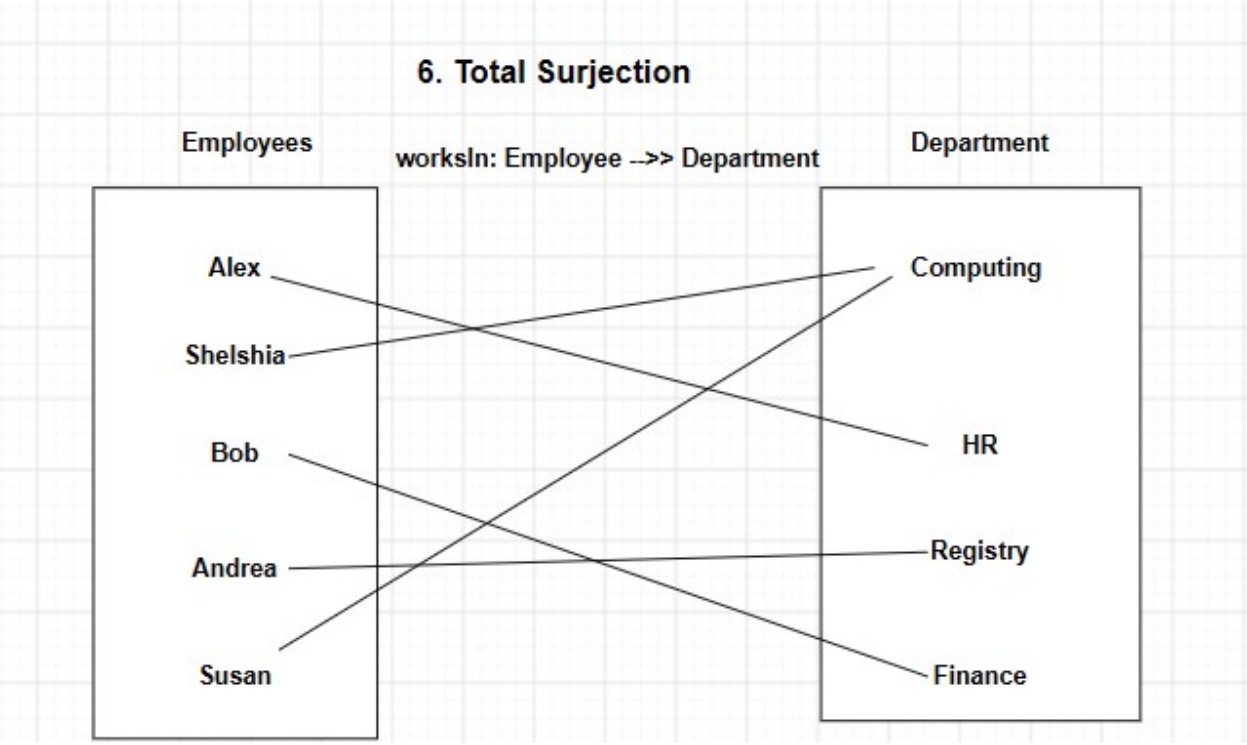


3. Total Surjective

For the function to be total function, all elements from source set should have a pair. And for it to be a surjective function, all elements of target set should have a pair.

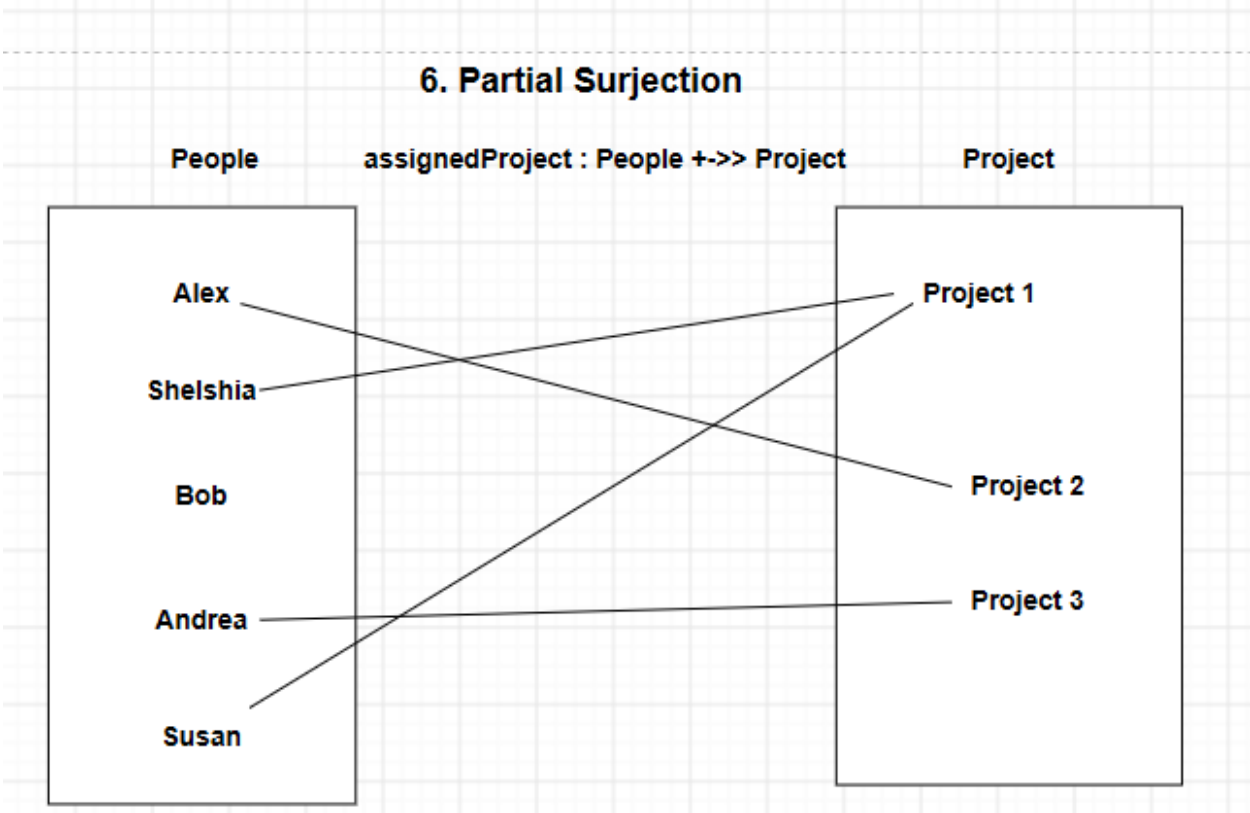
Ex – worksIn: Employee -- >> Department

Every employee will be assigned to at least and at most one department. Every department will have at least one Employee.



4. Partial Surjection

Not every element in source set has a pair and all the elements in target set will have a pair.



5. Total Bijection

hasStudentID: Student >->> IDs

Every student will have an id and there cannot be an id without assigning to student

6. Total Bijection

